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1. Find linear regression

$$\hat{\beta}_1 = \frac{S_{XY}}{S_{XX}} = \frac{\sum_{i=1}^{100} X_i Y_i - 100\bar{X}\bar{Y}}{\sum_{i=1}^{100} X_i^2 - 100\bar{X}^2} = \frac{186000 - 100\cdot 15\cdot 128}{24500 - 100\cdot 15^2} = \frac{-6000}{2000} = -3$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} = 128 - (-3)\cdot 15 = 173$$

$$\Rightarrow \hat{Y} = 173 - 3X$$

2. $H_0: \beta_1 = 0$ (無關), $H_1: \beta_1 \neq 0$, $\alpha = 0.05$

t-test : $\frac{\beta_1}{S_{\beta_1}} \sim t(100-2)$, $S_{\beta_1}^2 = \frac{MSE}{\sum_{i=1}^{100}(X_i - \bar{X})^2}$

$$MSE = \frac{SSE}{100-2} = \frac{\sum_{i=1}^{100}(Y_i - \beta_0 - \beta_1 X_i)^2}{98}$$

$$= \frac{\sum_{i=1}^{100} Y_i^2 + 100\cdot\beta_0^2 + \beta_1^2 \sum_{i=1}^{100} X_i^2 - 2\beta_0 \sum_{i=1}^{n} Y_i + 2\beta_0\beta_1 \sum_{i=1}^{n} X_i - 2\beta_1 \sum_{i=1}^{n} X_i Y_i}{98}$$

$$= \frac{1714000 + 2992900 + 220500 - 4428800 - 1557000 + 1116000}{98}$$

$$= \frac{57600}{98}$$

$$S_{\beta_1}^2 = \frac{MSE}{\sum_{i=1}^{n} X_i^2 - n\bar{X}^2} = \frac{576\cancel{00}}{98\cdot 200\cancel{0}} = \frac{144}{490}, \quad S_{\beta_1} = \frac{12}{7\sqrt{10}}$$

$$\frac{\beta_1}{S_{\beta_1}} = \frac{-\cancel{3}\cdot 7\sqrt{10}}{\cancel{12}\,4} = -\frac{7}{4}\sqrt{10} < -t_{0.025}(98)$$

$\Rightarrow$ reject H_0, accept $\beta_1 \neq 0$

3. $X = 15$, find Y with 95% confidence level.

$\beta_0 + 15\beta_1 = 173 - 45 = 128$, $t_{0.025}(98) \doteq 1.98$

$$S_{pred}^2 = MSE\left[1 + \frac{1}{100}\frac{(20-15)^2}{\sum_{i=1}^{100}(X_i - 15)^2}\right] = \frac{57600}{98}\left[1 + \frac{1}{4000}\right] = \frac{576\cancel{00}\times 4001}{98\times 400\cancel{0}} \doteq 587.9$$

$S_{pred} \doteq 24.25$

$$C.I. = [128 - 1.98\times 24.25,\ 128 + 1.98\times 24.25] = [79.985,\ 176.015]$$

4. ANOVA

$SSR = SSTO - SSE = 18000$

$SSE = 57600$

$SSTO = \sum_{i=1}^{100} (Y_i - \bar{Y})^2 = 1714000 - 100 \cdot 128^2 = 75600$

$(F_{0.05}(1,98) \doteq 3.938)$

Source of Variation	SS	DoF	MS	F-value
Regression	18000	1	18000	30.625
Error	57600	98	$\frac{57600}{98}$	
Total	75600	99		